
1. Beautiful Naturally

Program Name: Beautiful.java

Input File: beautiful.dat

Judges Input File

```
13
Mary 13.99
Ruth 22.04
Anne 12.39
Taylor 18.34
Rose 32.12
Tammie 22.12
Lynn 12.12
Ellen 18.23
Liz 15.23
Meg 18.79
Sherry 21.00
Kelly 15.67
Sari 9.34
```

Judges Output to Screen

```
$231.38
```

2. Blobs

Program Name: Blobs.java

Input File: blobs.dat

Judges Input File

```
4
7 8 4
. . . . * . . .
***** . . . .
***** . . . .
***** . . . .
. . . . ** . .
. . ** . . . .
. . ** . . . .
. . ** . . . .
2 1
5 3
1 5
7 4
5 6 5
. . . ***
*** . . .
. . . ***
** . ***
** . ***
5 6
1 6
5 1
3 3
4 5
3 3 3
***
***
***
2 2
3 3
1 1
9 13 7
***** . ** . *****
***** . ** . *****
***** . ** . *****
***** . ** . . . .
. . . . . ** . . . .
. . . . . **
***** . ** . . . .
***** . ** . . . .
***** . . . . . **
***** . ***** . **
4 3
7 8
7 10
9 10
8 5
3 6
5 13
```

Judges Output to Screen (see next page)

2. Blobs (cont.)

Judges Output to Screen

```
2 1
NO BLOB
1 5
6 3
3 4
1 4
4 1
NO BLOB
3 4
1 1
1 1
1 1
1 1
1 7
NO BLOB
9 6
NO BLOB
NO BLOB
5 11
```

3. Diamonds are a Girl's Best Friend

Program Name: Diamonds.java

Input File: diamonds.dat

Judges Input File

```
5
U 11
I 3
L 7
M 9
E 15
```

Judges Output to Screen

```
      U
     U U
    U   U
   U     U
  U       U
 U         U
U           U
 U         U
  U       U
   U     U
    U   U
     U U
      U
```

```
  I
 I I
  I
```

```
    L
   L L
  L   L
 L     L
L       L
 L     L
  L   L
   L L
    L
```

```
    M
   M M
  M   M
 M     M
M       M
 M     M
  M   M
   M M
    M
```

(continued on next page)

3. Diamonds are a Girl's Best Friend (cont.)

A diamond shape composed of the letter 'E'. The shape is 11 rows high and 11 characters wide. The top row has 1 'E', the second row has 2 'E's, and so on, until the middle row (row 6) which has 11 'E's. The bottom row has 1 'E', the row above it has 2 'E's, and so on, mirroring the top half. The overall shape is a diamond with a point at the top and bottom.

```
  E
 E E
E   E
 E   E
E   E
 E   E
E   E
 E   E
E   E
 E   E
  E
```

4. Elevator

Program Name: Elevator.java **Input File:** elevator.dat

Judges Input File

```
8
150 270 120 178 302 245 230 124 145 267 232 125 245 98 110 237
323 232 134 232 342 231 145 176 156 238 214 125
231 324 324 265 215 324 321 148 153 131
145 231 243 261 38 63
234 453 324 124 342 123 321 253 222
25 35 321 123 234 256 267 211 233 135 123
125 124 123 222 221 225 256 227 114 225 35 32 100 98
50 51 52 53 54 55 156 157 158 159 160 95 200 200 200 200
```

Judges Output to Screen

```
9
9
6
6
7
11
12
16
```

5. Hickory Dickory Dock

Program Name: Hickory.java

Input File: hickory.dat

Judges Input File

```
HICKORY DICKORY DOCK,  
THE MOUSE RAN UP THE CLOCK.  
THE CLOCK STRUCK ONE,  
THE MOUSE RAN DOWN,  
HICKORY DICKORY DOCK.  
NURSERY RHYMES ARE THE BEST  
WHY, I HAVE NO CLUE.  
THEY GIVE ME A REST.  
THEY MAKE ME NOT BLUE.  
THEY ARE NOT EVEN A TEST.
```

Judges Output to Screen

```
HICKORY DICKORY DOCK,  
THE CLOCK STRUCK ONE,  
HICKORY DICKORY DOCK.  
WHY, I HAVE NO CLUE.  
THEY MAKE ME NOT BLUE.
```

6. Life vs. School

Program Name: Life.java Input File: life.dat

Judges Input File

8

```
08:45 11:15
06:45 08:00 12:00 03:00
10:30 13:30
08:30 10:30 13:45 14:45 16:45 18:46
14:00 17:00
10:45 11:45 15:00 16:00 16:45 18:15
10:00 15:00
08:50 10:00 12:00 14:00 16:00 17:00
08:00 17:00
07:00 18:00
08:00 17:00
09:00 11:00
09:45 12:45
09:00 10:00 11:00 12:00 12:45 03:00
17:00 19:00
00:01 06:02 09:30 12:00 13:12 16:59
```

Judges Output to Screen

```
DAY #2 CONFLICT WITH CLASS #1
DAY #3 CONFLICT WITH CLASS #2
DAY #3 CONFLICT WITH CLASS #3
DAY #4 CONFLICT WITH CLASS #1
DAY #4 CONFLICT WITH CLASS #2
DAY #5 CONFLICT WITH CLASS #1
DAY #6 CONFLICT WITH CLASS #1
DAY #7 CONFLICT WITH CLASS #1
DAY #7 CONFLICT WITH CLASS #2
DAY #7 CONFLICT WITH CLASS #3
```

7. Miles Per Gallon Calculator

Program Name: MPGCalc.java

Input File: mpgcalc.dat

Judges Input File

```
11365
11415 2.7 HIGHWAY
11545 10.9 CITY
11650 9.3 CITY
11800 12.2 HIGHWAY_TRAFFIC
11843 2.9 HIGHWAY
11900 8.4 HIGHWAY
12000 10.0 CITY
12100 10.0 SNOW
12200 10.0 SNOW
```

Judges Output to Screen

CITY	11.1
HIGHWAY	10.7
HIGHWAY_TRAFFIC	12.3
SNOW	10.0

8. Palindromic Essays

Program Name: Palindrome.java

Input File: palindrome.dat

Judges Input File

5

BOB WAS FROM WASSAMASSAW, SOUTH CAROLINA. HE LOVED TO DRIVE HIS RACECAR. MARY WAS FROM KINIKINIK, COLORADO, AND SHE LOVED TO KAYAK ON THE RIVERS. THEY BOTH LOVED TO PLAY TIDDLYWINKS.

-

OTTO GOT A SUMMER JOB TO REPAPER THE DINING ROOM OF A HOUSE. THIS WAS HIS FIRST SUMMER JOB EVER. IT WAS HARD FOR OTTO TO START THE WALLPAPER LEVEL TO THE FLOOR BUT WITH PRACTICE, HE SUCCEEDED. HE GAVE THE MONEY HE EARNED TO HIS MOM.

-

ATOYOTA.

-

MADAM HANNAH RADAR MADAM HANNAH RADAR MADAM HANNAH RADAR MADAM HANNAH RADAR
MADAM HANNAH RADAR
MADAM HANNAH RADAR MADAM HANNAH RADAR
MADAM HANNAH RADAR

-

I DEIFIED, DELEVELED, AND DETARTRATED WHILE I DEVOVED AND DEWED MY IN THE WATER. RACECAR.
DETARTRATED.
DELEVELED DEVOVED

-

Judges Output to Screen

ESSAY #1:

BOB
KAYAK
KINIKINIK
RACECAR
WASSAMASSAW

ESSAY #2:

LEVEL
MOM
OTTO (2)
REPAPER

ESSAY #3:

ATOYOTA

ESSAY #4:

HANNAH (8)
MADAM (8)
RADAR (8)

ESSAY #5:

DEIFIED
DELEVELED (2)
DETARTRATED (2)
DEVOVED (2)
DEWED
RACECAR

9. Profit

Program Name: Profit.java Input File: profit.dat

Judges Input File

```
5
ABCDEFGHIJ Jerry 3
DECJ $39.00
ABC $2.01
CFIF $32.79
GUSOCPRTWD Tom 2
PTCS0 $679.35
WUG $10.29
QWERTYUIOP Mary 5
QW $1.00
TRWY $53.21
YTUR $98.99
POITU $1400.50
TYTY $65.65
ASDFGHJKLZ Robert 4
JFSFD $827.99
JSLDF $929.99
DDDD $38.79
ZZZZ $100.00
MNBVJHGFTR Ricky 5
GHFB $88.45
VBFGH $521.34
BGTVFR $3215.99
GHJT $100.55
GBHNG $1111.11
```

Judges Output to Screen

```
Jerry $13.54
Tom $107.31
Mary $488.65
Robert $525.01
Ricky $1269.64
```

10. Shake

Program Name: Shake.java **Input File:** shake.dat

Judges Input File

7

```
VI LO COMERPUTESCIE ANCE*  
  WINNINGIL DISTU RICT ISHE FIRSTTTT STEP WARD STODATE*ABCLEFGHIJK  
HYES T BESTEG THIN IN LSRIFE A FREEE  
ARICHD ISRB MY F*ABFGCDEF  
DMARY HAA LITTL IE LAMB S FLEECTIE AS WHE AS SNTEOW*ABCDGHIJKLMF  
  AUSTINS THE ITPLACE  BE INOB MAY.  THEREEE!*ABCD  
SAUINT1 #
```

Judges Output to Screen

```
I LOVE COMPUTER SCIENCE  
WINNING UIL DISTRICT IS THE FIRST STEP TOWARD STATE  
YES THE BEST THINGS IN LIFE ARE FREE  
RICHARD IS MY BFF  
MARY HAD A LITTLE LAMB ITS FLEECE AS WHITE AS SNOW  
AUSTIN IS THE PLACE TO BE IN MAY. BE THERE!  
AUSTIN #1
```

11. Shopping to the Max

Program Name: Shopping.java

Input File: shopping.dat

Judges Input File

7

14.99 79.99 80.29 13.44 15.17 33.29

88.10 15.00 111.90 45.14 2.25 33.45

20.00 50.00 10.00 20.00 30.00 40.00 70.01 30.10 25.25

122.55

16.00 56.10 43.90 35.18 38.04 21.09 48.31

99.01 99.02 50.00 40.01 49.99 18.25

33.24 34.25 35.26 36.27 37.28 38.29 39.30 13.39 12.23 11.12

Judges Output to Screen

6.43

.00

4.64

77.45

.51

.99

.21

12. Students' Grades

Program Name: Students.java

Input File: students.dat

Judges Input File

```
29
GEORGE 67
AMOS 54
ROBERT 87
RICHARD 98
MARY 45
ANN 98
LUCY 20
WENDY 64
SHARLA 62
JOHN 59
JOE DON 98
MARY LOU 90
LARRY 92
NAN 74
JAN 0
JERRY 69
MARY 100
ALICE 70
AL 39
JOE 40
JILL 99
KEN 6
KIT CARSON 12
KARLA 80
RUSS 59
ROD 60
IAN 80
SHAUN 50
SHAWN 77
```

Judges Output to Screen

```
<40: *****
40-49: **
50-59: ****
60-69: *****
70-79: ***
80-89: ***
90-100: *****
```

Note:

- Colons should be aligned vertically
- Labels should be right justified before the colon
- Spaces to the left of the labels are insignificant